



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
SECOND SESSIONAL
SUBJECT: (CE-313) DATABASE MANAGEMENT SYSTEMS

Examination : B.Tech Semester - III
Date : 06/09/2022
Time : 10:45 a.m. to 12:00 p.m.

Seat No. CE052
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

- Q.1 Do as directed.** [12]
- CO2** A (a) Examine the statement with example: "Functional dependencies help to identify repetition in a relation." [2]
- CO2** N (b) Identify the number of potential candidate keys possible for a relation with 3 attributes (A, B, C). State an example. [2]
- CO2** R (c) Describe the problems that could arise if a relation has transitive dependency. [2]
- CO3** U (d) Differentiate Ordered file organization and Unordered file organization. [2]
- CO1** A (e) Write a SQL statement to Select the orders placed by client 'Ivan', without using SQL joins. The required information is given in Figure 2. [2]
- CO3** R (f) Create a multitable clustering file structure for the relations given below. [2]

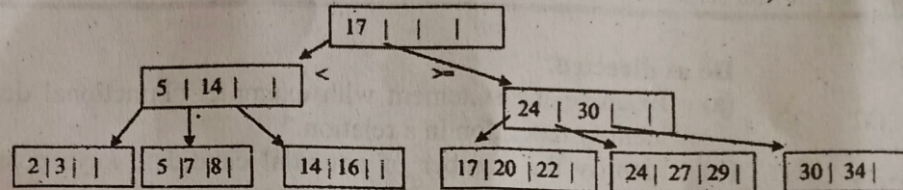
Student				Department	
S_Id	S_Name	Address	D_Id	D_Id	D_Name
1	Tanay	Delhi	4	0	Computer Engineering
2	Anay	Assam	2	1	History
3	Malay	Gujarat	5	2	Sociology
4	Vinay	Tamilnadu	1	3	Zoology
5	Sarita	Ladakh	4	4	Forensic Science
6	Parita	Tripura	3	5	Arts
7	Kavita	Maharashtra	0		
8	Nivedita	Madhyapradesh	2		

- Q.2 Attempt Any TWO from the following questions.** [12]
- CO2** E (a) What is the need of normalization? Consider Relation R having three attributes: A, B and C. The functional dependencies are given as $A \rightarrow B$ and $B \rightarrow C$. Check for following and perform normalization if needed: [6]
1. Is the relation in 2NF? $\checkmark$
 2. Is the relation in 3NF? $\checkmark$
- a - ck*
bc - nonPrime
- CO2** E (b) What will be the problem if decomposition is not done in the proper manner? Consider Relation R with attributes A, B, C, D and E. The functional dependencies are given as $A \rightarrow B$, $BC \rightarrow E$ and $ED \rightarrow A$. R is decomposed in R1 having attributes A, B and R2 having attributes A, C, D, E. Find whether the given decomposition is lossless or not. [6]
- CO2** E (c) What is the closure of the attribute set? Support the answer by describing the algorithm and suitable example. Mention the applications of the algorithm. [6]

- Q.3** Attempt the following questions. [12]
- CO3 N** (a) Suppose that we have an ordered file of 20,000 records and these records are stored on a disk and block size is 1024 bytes files records of fixed length and unspanned of size 100 bytes and suppose that we have created a secondary index on key field of size 9 bytes and a block pointer of size 6 bytes then find the average number of block access required with and without indexing. [6]
- CO3 C** (b) Construct B+ tree for 1,4,7,10,17,21,31,25,19,20,28,42,3 with P=4 (Order is 4). [6]

OR

- Q.3** Attempt the following questions. [12]
- CO3 C** (a) For the following B+-tree (order, n = 5) show the form of the tree after each of the of operations of the sequence: Delete 17; Delete 20; Delete 34 [4]



- CO1 N** (b) I. Draw a Dense index and Sparse index on BUILDING for records given in Figure 1. [2]
- II. Should we create a sparse index on non-clustering index? Justify your answer. [2]
- CO3 U** (c) Consider a file of departments' records for the university database. Each record of this file is defined as: department(ID varchar2(5), name varchar2(20), building_name varchar2(20), budget(10,2)) [4]
- The departments' records are represented as variable-length records. List the problems, must be solved by any technique implementing variable-length records. Explain the solution for each problem listed with the example of department record if suitable.

DEPT_NAME	BUILDING	BUDGET
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000

Figure 1

Name	Null?	Type
CLIENT_NO	NOT NULL	VARCHAR2(10)
NAME		VARCHAR2(20)
ADDRESS1		VARCHAR2(30)
ADDRESS2		VARCHAR2(30)
CITY		VARCHAR2(15)
STATE		VARCHAR2(15)
PINCODE		NUMBER(6)
BAL_DUE		NUMBER(10,2)
PHONE_NO		NUMBER(10)

CLIENT_MASTER

Name	Null?	Type
S_ORDER_NO	NOT NULL	VARCHAR2(6)
ORDER_DATE	NOT NULL	DATE
CLIENT_NO		VARCHAR2(6)
DELY_ADD		VARCHAR2(25)
SALESMAN_NO		VARCHAR2(6)
DELY_TYPE		CHAR(1)
BILL_YN		CHAR(1)
DELY_DATE		DATE
ORDER_STATUS		VARCHAR2(10)

SALES_ORDER

Figure 2